

NON-RECIPROCALITY In FERRITE-LESS DEVICES: A Way To Full Duplex Mobile Communication

Madhav Khindri is B.Tech in electronics and communication, having interests in robotics and embedded systems

5G standards are currently being laid out. It is estimated that 5G paradigm will be realised in 2020 with fully-set standards and entry of new devices in the market based on those standards. One such prospect is full duplex communication. With full duplex realisation, a single antenna could be shared between the transmitter and the receiver.

However, there is a fundamental physical precept called reciprocity, which does not allow sharing the antenna between the two. The only way to break reciprocity is the traditional way of using special types of materials known as ferrites.

Ferrites have particular applications in ferrite devices (gyrators, circulators, isolators, etc). These use their property of non-reciprocity. When two circularly polarised waves—one rotating clockwise and another anti-clockwise—are made to propagate through ferrite, these react differently to both waves by showing different permeabilities to the waves. For example, ϵ_{r1} , μ_{r1} and ρ_1 are for the left circularly polarised wave, and ϵ_{r2} , μ_{r2} and ρ_2 for

the right circularly polarised wave. (Here, ϵ stands for permittivity, μ for magnetic permeability and ρ for charge density.)

Non-reciprocal materials are mainly realised by magnetic materials through Faraday effect. Some non-reciprocal devices—used in most modern-day communication—use a circulator. A circulator is a microwave device in which each terminal is connected only to the next clockwise terminal. For example, port 1 is connected to port 2 only (and not to port 3), and port 2 is connected only to port 3 (not to port 4). Although, there is no restriction on the number of ports, three or four ports are mostly used.

Traditional circulators are used in radar systems. The transmitter feeds the antenna, while received energy is directed to the receiver. A powerful radar transmitter is isolated from the sensitive receiver. The same antenna can be used for both transmission and reception. This is duplexer action being performed by the circulator.

Use of a single antenna for both transmitter and receiver nodes using a conventional circulator is based on magnetic biasing of rare-earth ferrite materials. These are bulky, expensive and incompatible with standard integrated circuit (IC) technologies.

To remove the magnet, an active circulator relying on intrinsic non-reciprocal prop-

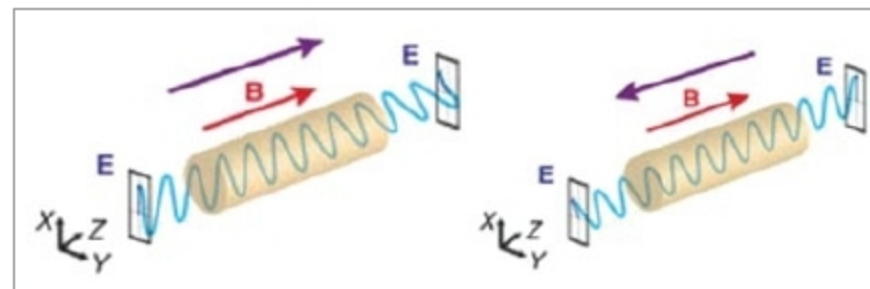


Fig. 1: Rotation of electromagnetic waves on magnetic field in forward and backward directions; E=electric field and B=magnetic field (Credit: Nature Communications, April 2016)

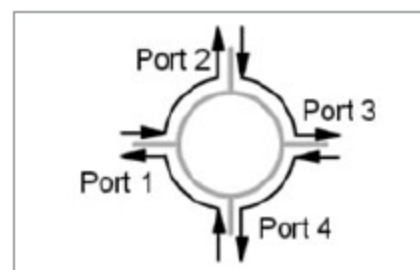


Fig. 2: A four-port circulator

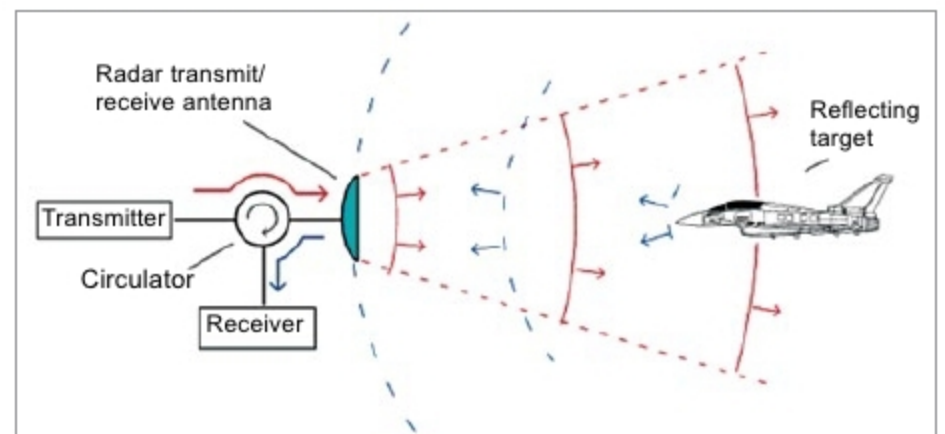


Fig. 3: Circulator in radar systems

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